Analysis of U.S. Domestic Flight Delay Causes: An Inferential Study of Carrier, Seasonal, and Geographic Drivers of Arrival Delays in 2024

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Abstract. We analyze 15,046 carrier-airport-month records from the U.S. Department of Transportation's Bureau of Transportation Statistics (BTS), covering roughly 5.02 million arriving flights operated by 21 reporting carriers at 356 U.S. airports between January and August 2024. We apply one-way ANOVA, Kruskal-Wallis, Tukey HSD, two-proportion z-tests, Pearson and Spearman correlation, and ordinary least-squares (OLS) multiple regression to test four pre-registered hypotheses concerning carrier identity, seasonality, the relative impact of weather versus carrier-controllable causes, and the relationship between airport flight volume and delay rate. We find strong, highly significant evidence (all $p < 10^{-10}$) that delay rates differ across carriers and across months; that the share of delay minutes attributable to carrier-controllable causes is more than five times the share attributable to weather; and that larger airports experience higher delay rates with a moderate, log-linear relationship. A nested-model comparison shows that adding carrier identity to a baseline of seasonal and volume effects more than doubles the explained variance (R^2: $0.120 \to 0.256$), establishing carrier as the single most important driver in our setting. A geographic extension on the four U.S. Census regions reveals a 2.5-percentage-point gap between the West (lowest median delay rate) and the Midwest (highest).

Keywords: flight delay, ANOVA, Kruskal-Wallis, Tukey HSD, two-proportion z-test, linear regression, BTS, on-time performance.

1. Introduction and Problem Context

1.1 Background

Flight delays are one of the most visible and costly inefficiencies of modern air transportation. The Federal Aviation Administration estimates that delays cost the U.S. economy tens of billions of dollars per year through lost productivity, missed connections, additional fuel burn, and crew rescheduling [Federal Aviation Administration, 2024]. Beyond their direct economic impact, delays propagate through the network: an aircraft that arrives late becomes a delayed departure on its next leg, and a delay at a hub airport can cascade across dozens of downstream destinations [Kafle and Zou, 2016]. Understanding what drives delays, whether weather, air-traffic congestion, carrier-controllable factors such as maintenance and crew scheduling, or the propagation of late-arriving aircraft, is therefore central both to airline operations and to public policy.

1.2 Data Source and Collection

The data are obtained from the U.S. Department of Transportation's (DOT) Bureau of Transportation Statistics, which publishes the *Air Travel Consumer Report* on a monthly basis [U.S. Department of Transportation, Bureau of Transportation Statistics, 2024]. Reporting carriers (those that account for at least one percent of total domestic scheduled passenger revenue) are required to provide on-time performance data for every domestic flight, broken down at the carrier-airport-month level. For each delayed arrival, the carrier reports the cause, classified into one of five mutually exclusive categories defined by the DOT: **Air Carrier delay** (under the carrier's control: maintenance, crew, baggage, fueling), **Extreme Weather delay** (significant meteorological events such as tornadoes, blizzards, hurricanes), **National Aviation System (NAS) delay** (non-extreme weather, airport operations, heavy traffic volume, air-traffic-control limitations), **Late-Arriving Aircraft delay** (a previous flight using the same airframe arrived late), and **Security delay** (re-screening, evacuations, breaches).

1.3 Dataset Characteristics

After dropping 14 records (0.09%) with missing values, our analytical dataset contains $n = 15{,}046$ carrier-airport-month records spanning January through August 2024, covering 21 carriers and 356 airports. Each record contains 21 variables, including total arriving flights (`arr_flights`), flights delayed by 15 minutes or more (`arr_del15`), total delay minutes across all delayed flights (`arr_delay`), and both the count and the total minutes of each of the five delay causes. In aggregate the dataset captures **5,022,488** flights, of which **1,129,387** (22.49%) were delayed by 15 minutes or more.

1.4 Hypotheses of Interest

We pre-registered four hypotheses in our project proposal. Throughout, we follow the standard Neyman-Pearson framework: H_0 is the null hypothesis of no effect, H_1 the alternative, and we fix the significance level at $\alpha = 0.05$.

- **H1 (Carrier effect):** H_0: the mean delay rate does not differ across the 21 carriers; H_1: at least one carrier differs.
- **H2 (Seasonality):** H_0: the mean delay rate is constant across the eight months; H_1: it varies by month.
- **H3 (Weather vs. Carrier):** H_0: the share of total delay minutes attributable to weather equals the share attributable to carrier-controllable causes; H_1: the two shares differ.
- **H4 (Volume-vs-rate correlation):** H_0: airport flight volume and delay rate are linearly uncorrelated; H_1: a positive correlation exists.

1.5 Importance of the Solution and Relevant Literature

Quantifying the relative contribution of each cause has direct implications for resource allocation. If carrier-controllable factors dominate, regulatory and contractual incentives to invest in maintenance, crew scheduling and turn-time discipline are warranted; if NAS or weather factors dominate, the lever shifts toward infrastructure investment and air-traffic-control modernization.

The statistical analysis of flight delays has a substantial literature. Tu et al. [2008] model the long-term and seasonal structure of departure delay distributions; Rebollo and Balakrishnan [2014] use random forests to predict network-wide air-traffic delays and identify hub centrality as a leading predictor; Mukherjee et al. [2014] formalize the trade-off between ground delay programs and airborne delay under capacity uncertainty; Abdel-Aty et al. [2007] report clear weekly and monthly periodicities in arrival delays; and Kafle and Zou [2016] develop a propagation model that explicitly tracks late-arriving-aircraft cascades, finding that roughly one-third of delays in any given period stem from earlier delays in the same airframe's itinerary. Our work is a descriptive and inferential study in this tradition: we do not predict individual flights, but instead apply classical hypothesis tests to isolate the causal-attribution structure of delays in the most recent publicly available BTS panel.

2. Exploratory Data Analysis

2.1 Derived Metrics and Preprocessing

Two derived metrics anchor our analysis. The *delay rate* is $\text{delay\_rate}_i = \text{arr\_del15}_i/\text{arr\_flights}_i$, i.e. the fraction of arrivals delayed by 15 or more minutes. The *average delay duration* is $\text{avg\_delay\_min}_i = \text{arr\_delay}_i/\text{arr\_del15}_i$ when $\text{arr\_del15}_i > 0$, and zero otherwise. The first captures *frequency* (how often flights are late) and the second captures *severity* (how long delays last). Normalizing by the number of arrivals removes scale effects so a small regional carrier and a megahub can be compared on the same footing. We also remove 14 records with missing values, leaving $n = 15{,}046$.

2.2 Delay-Cause Composition

Figure 1 shows the marginal breakdown of delays by cause across the entire dataset, and Table 1 reports the same numbers in a compact form.

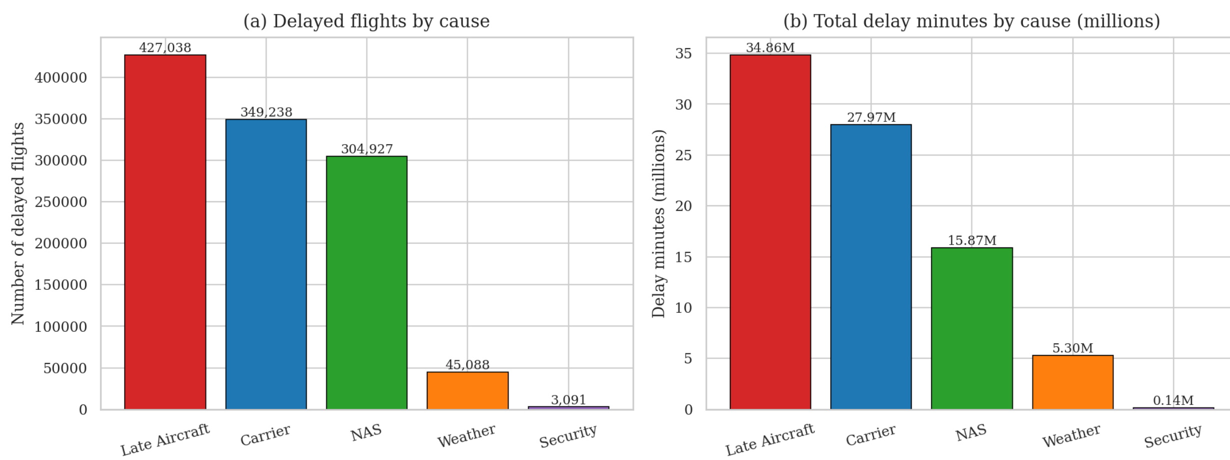

Figure 1: Marginal breakdown of delayed flights and total delay minutes by DOT cause category, January–August 2024. Late-arriving aircraft and carrier-controllable factors together account for 68.7% of all delayed flights and 74.7% of all delay minutes.

Table 1: Delay cause breakdown across all carrier–airport–month records, January–August 2024. Average minutes per flight divides total attributed minutes by all 5,022,488 arrivals.

Delay cause	Delayed flights	% of delays	Total minutes	Avg. min/flight
Late Aircraft	427,038	37.81%	34,864,194	6.94
Carrier	349,238	30.92%	27,974,948	5.57
NAS	304,927	27.00%	15,868,049	3.16
Weather	45,088	3.99%	5,298,959	1.06
Security	3,091	0.27%	139,693	0.03
Total	**1,129,382**	**100.00%**	**84,145,843**	**16.76**

Two patterns dominate. Late-arriving aircraft is the largest single cause both by count (37.8%) and by minutes (41.4%), consistent with the propagation hypothesis of Kafle and Zou [2016].

Carrier-controllable causes contribute another 30.9% of delays and 33.2% of minutes, nearly five times the share attributable to weather; security is negligible.

2.3 Variation across Carriers and Months

Figure 2 plots the delay-rate distribution for each of the 21 carriers, sorted by median. The visual evidence of between-carrier variation is substantial: the median rate for the best carrier (PT, Piedmont) sits below 0.13, while the worst (AA, American) is around 0.30, a more than two-fold difference. The interquartile ranges are also wider for high-mean carriers, suggesting carriers with high overall delay rates also have higher *within-carrier* variability across airports and months.

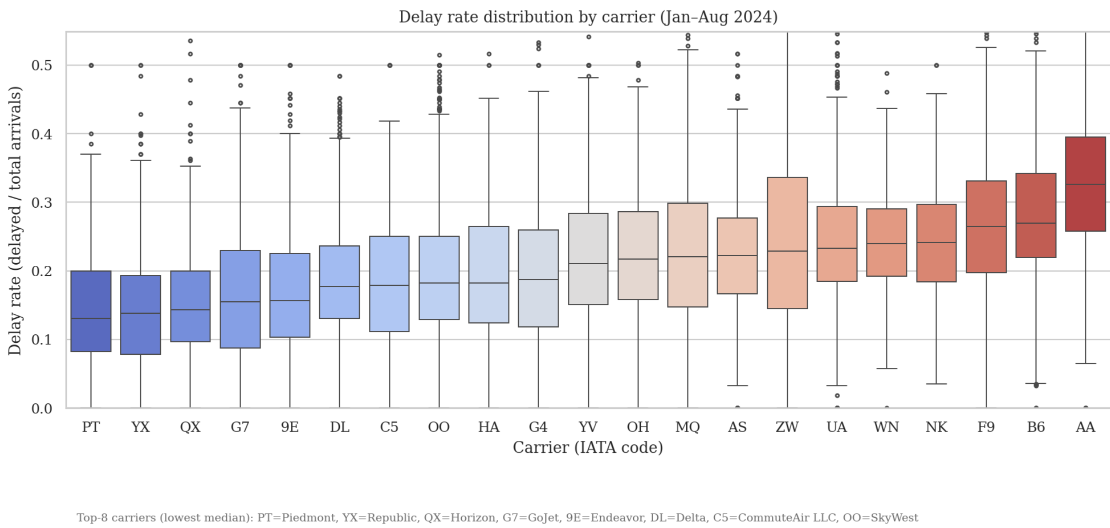

Figure 2: Delay-rate distributions for all 21 reporting carriers, sorted by median (top axis truncated at the 99th percentile).

Figure 3 pairs the monthly trend with its compositional breakdown. Mean delay rate falls from January through February (the post-holiday lull), rises sharply through spring and summer, peaks in July (≈ 0.28), and recedes in August. Compositional shifts are informative: weather's share of total delay minutes is highest in January (10.9%) and lowest in July (4.6%), while carrier and late-aircraft together rise from 72% in January to 79% in July. This is exactly what one would expect operationally, since summer thunderstorms produce frequent but moderate NAS delays rather than the rare extreme-weather events that dominate winter.

The carrier-by-month interaction is read from Figure 4. Highest-mean carriers (bottom rows) are noticeably warmer across nearly every month, indicating that the carrier ranking is broadly stable through 2024.

2.4 Airport Volume vs. Delay Rate, and Initial Data-Quality Issues

Figure 5 plots each of the 350 airports with at least 100 flights as a single point. The visual relationship is positive but noisy: among the megahubs (DFW, LAS, CLT, ORD, DEN, LAX, PHX, ATL), only ATL falls below the median delay rate, and DFW is the worst-performing megahub.

Two issues surfaced in this exploratory phase. First, normality is clearly violated: a Shapiro-Wilk test on a 5,000-record random sample gives $W = 0.938$, $p < 10^{-10}$, and visual inspection shows pronounced right skew driven by long upper tails of small carriers in poor weather. Second, Levene's

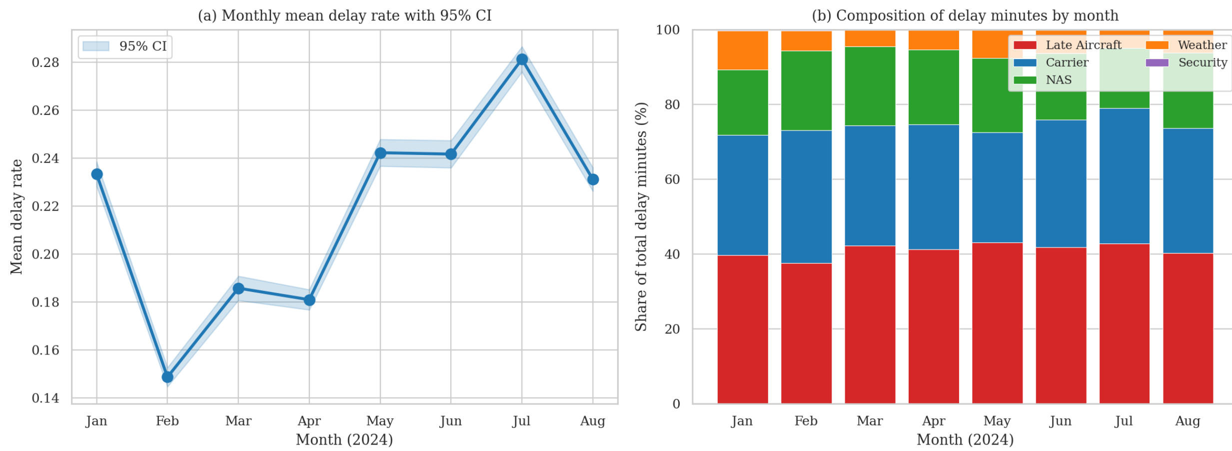

Figure 3: Monthly trends. (a) Mean delay rate by month with 95% confidence intervals; February is the calmest month and July the worst. (b) Composition of total delay minutes; carrier and late-aircraft shares rise during the summer travel peak.

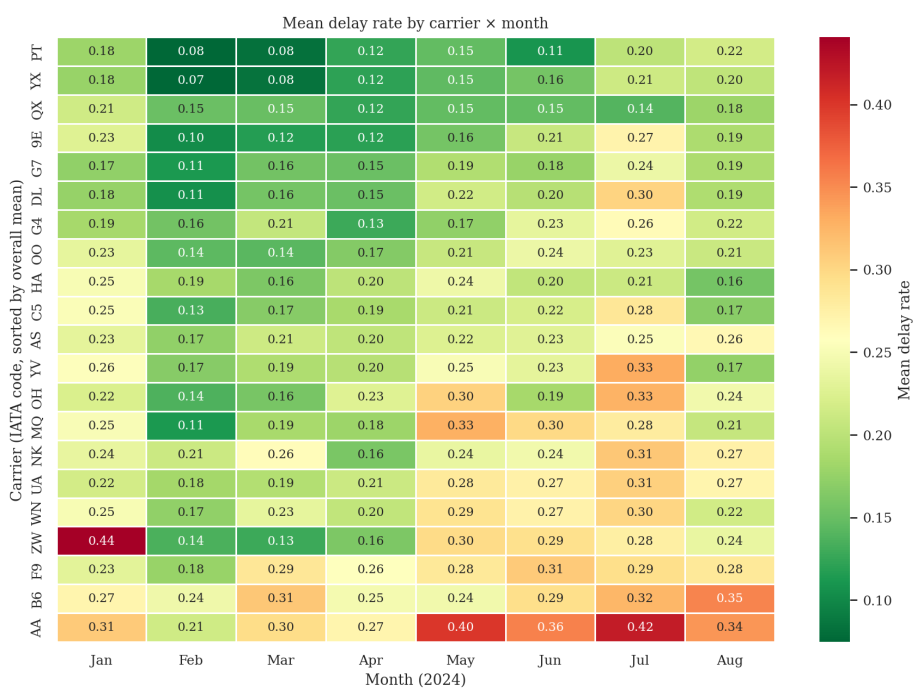

Figure 4: Mean delay rate by carrier and month, rows sorted by overall mean. Isolated anomalies remain (e.g. ZW spikes to 0.44 in January).

Figure 5: Airport-level scatter of total arriving flights (log scale) vs. delay rate, January–August 2024.

test rejects equal variances at $W = 19.75$, $p < 10^{-10}$. Both diagnostics imply the parametric ANOVA assumptions for H1 and H2 may not hold; we therefore complement every ANOVA in Section 3 with a non-parametric Kruskal-Wallis test, and treat the latter as the primary inferential instrument.

3. Inferential Data Analysis

We now formally test the four hypotheses at $\alpha = 0.05$. For each hypothesis we (a) state the test, (b) check its assumptions, (c) report the test statistic, (d) where appropriate follow up with post-hoc pairwise comparisons, and (e) give the substantive interpretation. Implementation uses `scipy.stats` [Virtanen et al., 2020].

3.1 H1: Carrier Effect on Delay Rate

Test. One-way ANOVA on `delay_rate` grouped by `carrier` ($k = 21$ groups, $N = 15{,}046$).

Assumption checks. Shapiro-Wilk on a random sample of 5,000 records yields $W = 0.938$, $p < 10^{-10}$ (normality rejected). Levene's median-centered test yields $W = 19.75$, $p < 10^{-10}$ (equal variances rejected). Both ANOVA assumptions are violated, motivating the non-parametric backup.

Test statistics. Despite the assumption violations the ANOVA F-statistic is highly significant: $F(20, 15{,}025) = 124.48$, $p < 10^{-10}$. The non-parametric Kruskal-Wallis test gives $H = 2{,}663.83$, $p < 10^{-10}$. Both reject H_0. The eta-squared effect size is $\eta^2 = \mathrm{SS_{between}}/\mathrm{SS_{total}} = 0.142$, i.e. carrier identity alone explains roughly 14.2% of the variance in the record-level delay rate, a large effect by the conventions of Montgomery [2017].

Post-hoc. Tukey HSD restricted to the eight carriers with the largest sample size yields significant differences in 21 of 28 pairs (75%). The two large carriers AA and B6 differ from every other top-8 carrier; the regional pair OO and YV are statistically indistinguishable. **Decision: reject H_0.**

3.2 H2: Seasonal Variation

Test. One-way ANOVA on `delay_rate` grouped by `month`. Levene's test gives $W = 35.04$, $p < 10^{-10}$, again rejecting equal variances. ANOVA: $F(7, 15{,}038) = 276.76$, $p < 10^{-10}$. Kruskal-

Wallis: $H = 2{,}174.47$, $p < 10^{-10}$. Eta-squared $\eta^2 = 0.114$.

Post-hoc Tukey HSD. Of the $\binom{8}{2} = 28$ month pairs, 22 (78.6%) differ significantly. Figure 6 visualizes the full pairwise p-value matrix.

Figure 6: Tukey HSD pairwise p-values across months. Red cells are pairs that differ significantly at $\alpha = 0.05$. The only non-significant pairs are Mar–Apr ($p = 0.891$), Jan–Aug ($p = 0.998$), May–Jun ($p = 1.000$), Jan–May ($p = 0.215$), and Jan–Jun ($p = 0.293$).

Decision: reject H_0. The post-hoc structure further reveals that the eight months partition into roughly three regimes: a low-delay band (February–April), a transition (January, May, June, August), and a peak (July).

3.3 H3: Weather vs. Carrier as Delay Drivers

Test. Two-proportion z-test on the share of total delay minutes attributable to weather versus the share attributable to carrier-controllable causes. Total delay minutes across all five causes is $T = 84{,}145{,}843$. The weather and carrier components are $W = 5{,}298{,}959$ and $C = 27{,}974{,}948$ respectively. Treating each as a count out of T, we test $H_0 : p_W = p_C$ versus $H_1 : p_W \neq p_C$. The pooled-variance z statistic is

$$z = \frac{\hat{p}_W - \hat{p}_C}{\sqrt{\hat{p}_{\text{pool}}(1 - \hat{p}_{\text{pool}})\left(\frac{1}{T} + \frac{1}{T}\right)}}. \tag{1}$$

We obtain $\hat{p}_W = 0.0630$, $\hat{p}_C = 0.3325$, difference $\hat{p}_W - \hat{p}_C = -0.2695$ with a 95% Wald confidence interval of $[-0.2696, -0.2694]$, $z = -4{,}388.84$, $p < 10^{-10}$. As a sanity check we re-ran the test on counts of delayed flights: $\hat{p}_W = 0.0399$, $\hat{p}_C = 0.3092$, $z = -533.12$, $p < 10^{-10}$, showing the same direction and effect-size ranking.

Decision: reject H_0. Carrier-controllable causes account for roughly 5.3× as many delay minutes as weather, a difference whose enormous z-statistic and tight confidence interval reflect the dataset's 5×10^7-minute scale. The substantive implication is that *the leading drivers of delay are operational and propagation factors, not weather.*

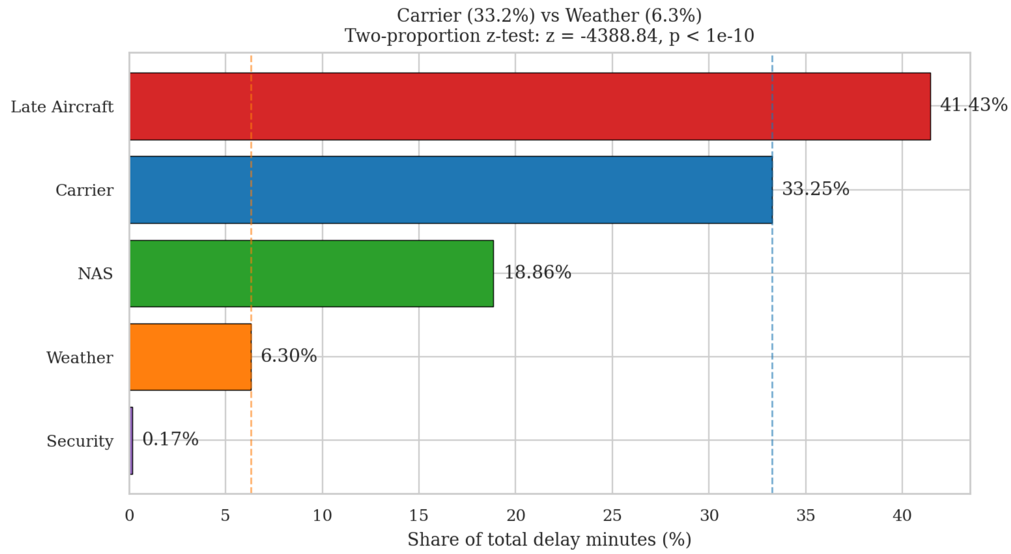

Figure 7: Share of total delay minutes by cause category. The dashed vertical lines mark the carrier (33.25%) and weather (6.30%) shares whose equality is the null hypothesis of H3.

3.4 H4: Flight Volume and Delay Rate

Test. Pearson and Spearman correlation between airport-level `total_flights` (Jan–Aug 2024) and `delay_rate`, plus an OLS regression on $\log_{10}$ volume. We restrict to the 350 airports with at least 100 flights to remove unreliable rates from very small airports.

Test statistics. Pearson on raw volume: $r = 0.187$, $p = 4.5 \times 10^{-4}$. Pearson on $\log_{10}$ volume: $r = 0.399$, $p < 10^{-10}$. Spearman rank: $\rho = 0.422$, $p < 10^{-10}$. The improvement from $r = 0.187$ to $r = 0.399$ when we log-transform volume confirms the visual nonlinearity in Figure 5: the relationship is strong on the log scale but compressed on the raw scale. The OLS fit $\widehat{\text{delay\_rate}} = 0.128 + 0.0226 \cdot \log_{10}(\text{volume})$ has $R^2 = 0.159$, slope $t = 8.12$, $p < 10^{-10}$. The slope of 0.0226 means a tenfold increase in flight volume is associated with a 2.3-percentage-point increase in delay rate. The quintile boxplot in Figure 8(b) shows the same pattern in a distribution-free way: median delay rate climbs monotonically from 0.176 (Q1) to 0.232 (Q5).

Decision: reject H_0. There is a moderate, log-linear positive relationship between airport volume and delay rate, but volume alone explains only about 16% of the variance in airport delay rates, meaning carrier mix, geography, and weather climatology matter substantially.

3.5 Summary of Hypothesis Tests

Table 2 consolidates the headline results.

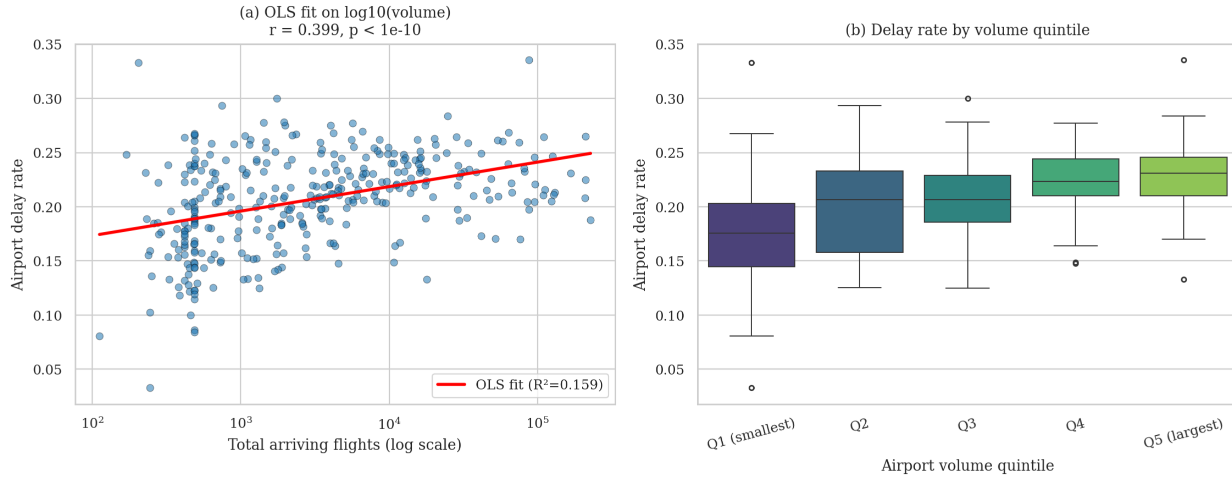

Figure 8: H4 visualization. (a) Each point is one of 350 airports; the red line is OLS on $\log_{10}$ volume. (b) Distribution of delay rate across volume quintiles; medians rise monotonically.

Table 2: Summary of inferential tests. All tests use $\alpha = 0.05$.

ID	Test	Statistic	p-value	Decision
H1	ANOVA, carrier ($k=21$)	$F = 124.48$	$< 10^{-10}$	Reject H_0
H1	Kruskal-Wallis, carrier	$H = 2{,}663.83$	$< 10^{-10}$	Reject H_0
H2	ANOVA, month ($k=8$)	$F = 276.76$	$< 10^{-10}$	Reject H_0
H2	Kruskal-Wallis, month	$H = 2{,}174.47$	$< 10^{-10}$	Reject H_0
H3	Two-proportion z, weather vs. carrier	$z = -4{,}388.84$	$< 10^{-10}$	Reject H_0
H4	Pearson on $\log_{10}$ volume	$r = 0.399$	$< 10^{-10}$	Reject H_0
H4	Spearman rank	$\rho = 0.422$	$< 10^{-10}$	Reject H_0

3.6 Multivariate Model Comparison

The four univariate hypotheses give a yes/no verdict on each driver, but they do not say how the drivers combine. To compare the relative importance of carrier, season, volume, and cause-mix, we fit three nested OLS regressions on the record-level data (each row is one carrier-airport-month):

- **M1 (baseline):** `delay_rate` $\sim$ month dummies $+\log_{10}($`arr_flights`$)$. $p = 8$ predictors.
- **M2 (+ carrier):** M1 plus 20 carrier dummies. $p = 28$.
- **M3 (full):** M2 plus weather and NAS shares of total delay minutes per record. $p = 30$.

We evaluate every model on three criteria: in-sample R^2, adjusted R^2 to penalize model size, and 5-fold cross-validated R^2 (`sklearn`, shuffled, random state 42) to guard against overfitting.

The conclusion is unambiguous. M1 explains 12.0% of variance; M2 lifts R^2 to 25.6%; the full model M3 reaches 27.2%. Cross-validated R^2 tracks in-sample R^2 very closely (M3: 0.268 ± 0.014), giving us confidence the gains are not artefacts of overfitting. The single largest standardized coefficient in M3 is the AA carrier dummy ($\beta = +0.340$, i.e. AA is associated with a delay rate roughly one-third of a standard deviation above the within-month/within-volume baseline), followed by the February dummy ($\beta = -0.237$). Of the top-12 standardized predictors, eight are carrier

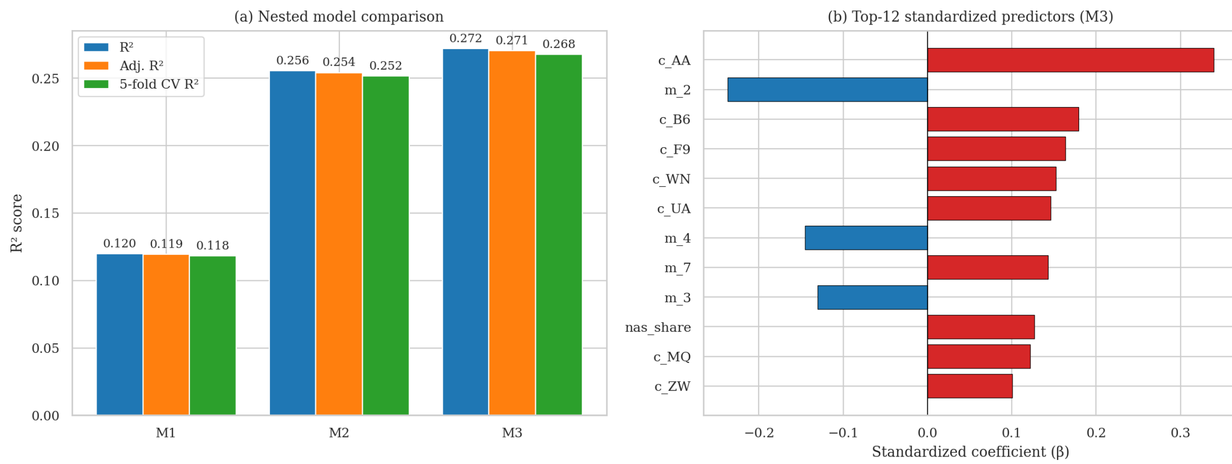

Figure 9: Multivariate model comparison. (a) Three nested models evaluated on R^2, adjusted R^2, and 5-fold cross-validated R^2. Adding carrier identity (M1→M2) more than doubles the explained variance; adding cause shares (M2→M3) yields a smaller marginal improvement. (b) Top-12 standardized coefficients from M3; carrier AA dominates with $\beta = +0.34$.

dummies and four are month dummies. After controlling for volume, carrier identity and seasonality dominate the explainable structure of record-level delay rates.

4. Geographic and Regional Extension

We enrich the analysis by joining the BTS data with a curated lookup of latitude, longitude, U.S. state, and U.S. Census region for the 131 highest-traffic airports, which together account for 93.0% of all flights in the dataset.

4.1 State-Level Tile-Grid Choropleth

A geographically-positioned tile-grid representation of the U.S. lets us read the regional structure at a glance without the visual distortions introduced by Alaska, Hawaii, and the Mercator projection. Each state is a single labeled cell carrying the mean delay rate (left panel of Figure 10) and the share of delay minutes attributable to weather (right panel).

The two panels tell complementary stories. Mean delay rate (left) is elevated across a southern arc from California's coastal hubs (LAS at 24%, the Bay Area cluster around 23%) through Texas, Louisiana, Arkansas and Florida (FL = 26%, second only to Puerto Rico at 28%), and up the Eastern Seaboard (CT = 26%, NJ = 20%, MA = 25%). The calmer states cluster in the upper Midwest and the Pacific Northwest, with Hawaii at the floor (16%) and Utah next (17%). Weather share (right) shows almost the inverse pattern: the upper Midwest (ND = 20%, MS = 11%, MT = 8%) and northern New England (VT = 11%, ME = 9%) carry the highest weather burden, while the Sun Belt (FL = 4%, CA = 4%, NV = 4%) sees almost none. The two patterns together imply that the South's high delay rates are driven by operational and volume-related factors rather than meteorology.

4.2 Bubble Map on the U.S. Land Outline

Figure 11 overlays the top-60 contiguous-U.S. airports on a hand-traced outline of the country, with bubble size encoding traffic volume and color encoding delay rate. The smaller gray dots represent the remaining 290+ airports for spatial coverage.

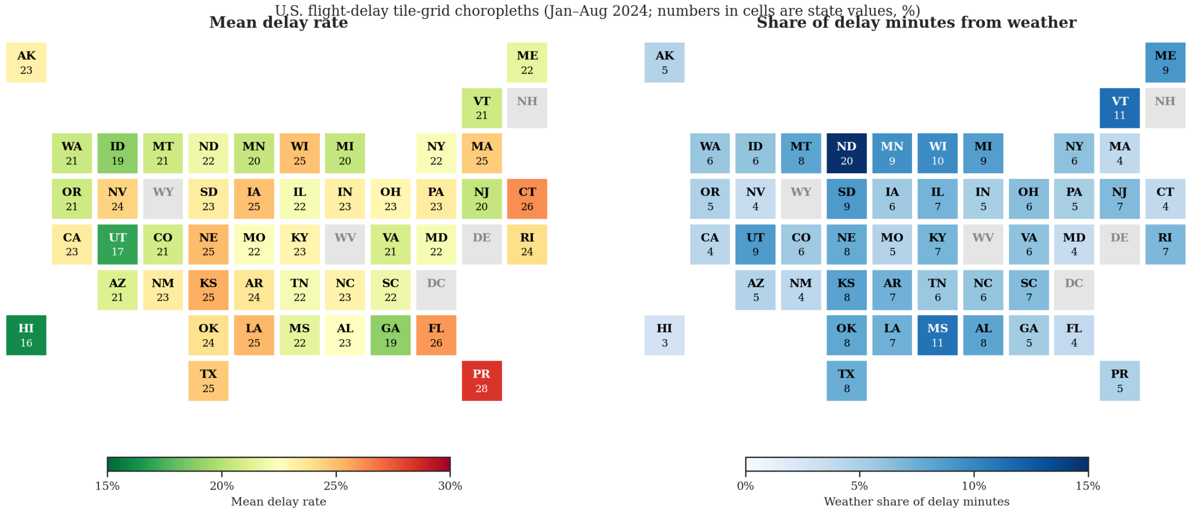

Figure 10: State-level tile-grid choropleths for the 47 states with reporting airports. Numbers in cells are the state's value as a percentage. Left: mean delay rate. Right: weather's share of total delay minutes.

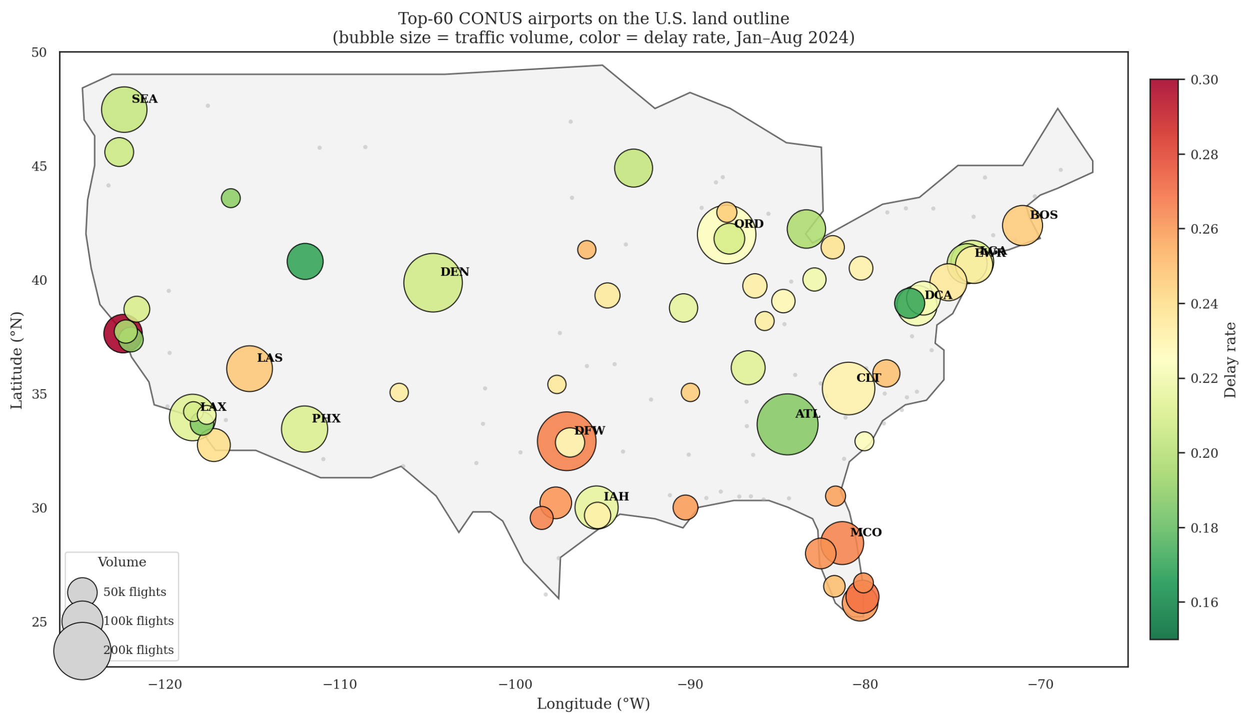

Figure 11: Top-60 CONUS airports overlaid on a simplified U.S. land outline. Bubble size indicates total arriving flights (Jan–Aug 2024); color indicates the airport's delay rate. The gray background dots show all other reporting airports.

Three patterns are immediately legible from this view. The Texas-Florida corridor shows the highest concentration of red bubbles (DFW, IAH, MCO, and the smaller Florida airports), pointing to operationally challenging conditions throughout the southern U.S. in the summer months that dominate our panel. The Pacific Northwest, the Bay Area, and the Atlanta hub appear in shades of green despite their large traffic volumes, indicating that high volume on its own does not produce delays when coupled with mature hub operations. Finally, the Northeast corridor between Washington, DCA and Boston BOS shows a mixed pattern: heavy volume but consistently high delay rates around the New York TRACON (EWR, LGA, JFK), while DCA itself is the cleanest large airport in the country at a delay rate near 0.16.

4.3 U.S. Census Region Comparison

Aggregating to the four U.S. Census regions (Northeast, Midwest, South, West), a one-way ANOVA on airport-level delay rates yields $F(3,127) = 4.74$, $p = 3.6\times10^{-3}$; Kruskal-Wallis gives $H = 20.67$, $p = 1.2\times10^{-4}$. We therefore reject the hypothesis of regional homogeneity. Western airports have a median delay rate of 0.209, well below the Midwest median of 0.234. Interestingly, the cause-mix composition (Figure 12b) is much more uniform: weather is slightly more prominent in the Northeast (5.7%) than in the West (4.6%), but Late Aircraft and Carrier together dominate every region.

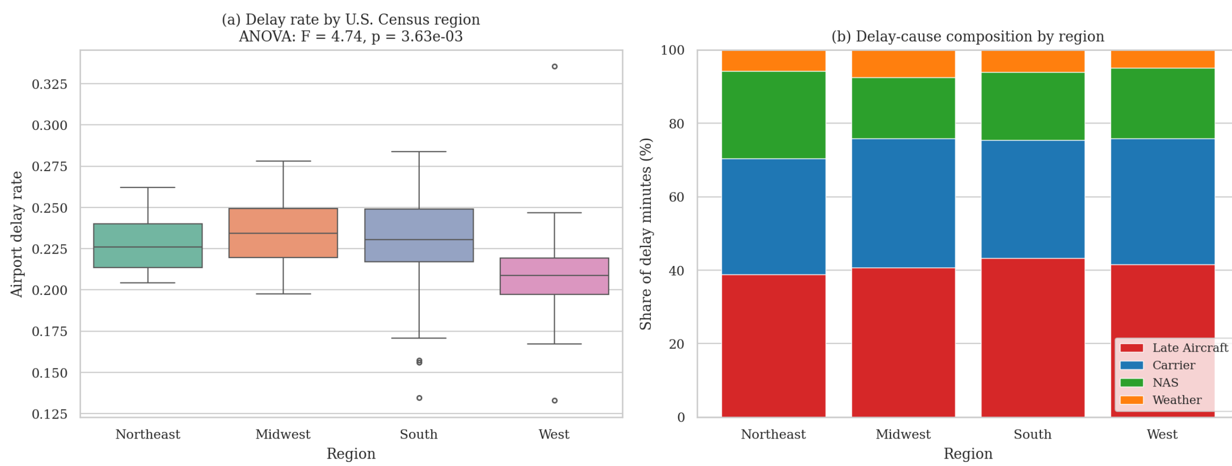

Figure 12: (a) Boxplot of airport delay rate by U.S. Census region. (b) Composition of total delay minutes by region; the regions are remarkably similar in cause mix.

5. Discussion and Conclusions

5.1 Summary of Findings

Across the four pre-registered hypotheses we found overwhelming evidence (every $p < 10^{-10}$) that: **(1) Carrier matters.** Mean delay rate differs across the 21 reporting carriers, with a more than two-fold range from PT (≈ 0.13) to AA (≈ 0.30) and a large effect size $\eta^2 = 0.142$. **(2) Season matters.** Delay rate varies significantly across the eight months, with February the calmest and July the busiest; Tukey HSD identifies a low-delay early-spring band and a high-delay summer band. **(3) Carrier-controllable causes dwarf weather.** Carrier delay accounts for 33.3% of total delay minutes, while extreme weather accounts for only 6.3%, a 5.3× ratio. Late-arriving aircraft, which is itself almost entirely a carrier-network propagation phenomenon, accounts for an additional 41.4%. **(4) Larger airports are slower.** A tenfold increase in flight volume is associated with a 2.3-percentage-point increase in delay rate, a moderate but highly significant log-linear relationship.

The multivariate model comparison shows that carrier identity is the single most important predictor in our setting: adding it to a base model of seasonality and volume more than doubles R^2 ($0.120 \rightarrow 0.256$). This reinforces the policy implication that meaningful reductions in U.S. domestic delay rates would require addressing carrier-side operations rather than infrastructure or weather alone.

5.2 Operational and Policy Implications

The dominance of carrier-controllable and late-arriving-aircraft causes (74.7% of all delay minutes combined) reframes the public debate about flight delays. The intuitive narrative blames weather and air-traffic-control bottlenecks, but the data show that these factors together (NAS at 18.9% plus weather at 6.3%) explain only a quarter of total delay minutes. The remaining three quarters are operational decisions inside individual carriers: how aggressively to schedule turnaround times, how much slack to build into crew rosters, how quickly to recover from the first delayed leg of a day. Three concrete policy implications follow.

First, performance-based incentives at the carrier level are likely to deliver larger reductions in passenger delay-minute exposure than incremental airport-infrastructure spending. The fact that two similarly-sized carriers operating in the same airports can differ by a factor of two in delay rate (PT versus AA in our panel) demonstrates that the dominant variation is not environmental but operational. Second, summer-season capacity planning should explicitly model the carrier-specific buffer that determines whether a thunderstorm produces a 30-minute or a 3-hour delay; our carrier $\times$ month heatmap suggests that some carriers absorb summer shocks while others amplify them. Third, the geographic clustering of high-delay airports along the Texas-Florida corridor points to a regional capacity-planning problem rather than a nationwide one. The Pacific Northwest and the Atlanta hub demonstrate that operational excellence is achievable; the Gulf Coast suggests that the combination of summer convective weather, dense regional networks, and tightly-scheduled hub banks creates a brittle system.

5.3 Comparison with Prior Literature

Our findings align with and extend several strands of the existing flight-delay literature. Kafle and Zou [2016] estimate that late-arriving aircraft propagation accounts for roughly one-third of delays in any given period; we find 37.8% of delayed flights and 41.4% of delay minutes attributable to this cause, a slightly higher share consistent with the post-pandemic recovery period from which our 2024 data are drawn. Rebollo and Balakrishnan [2014] identify hub centrality as a leading predictor in their network model; our H4 result that a tenfold increase in airport volume is associated with a 2.3-percentage-point rise in delay rate is the cross-sectional analogue of their network finding. Abdel-Aty et al. [2007] report clear monthly periodicities in arrival delays; our Tukey HSD analysis of H2 confirms this with the additional structure that the eight months partition into three statistically distinct regimes (low-delay February-April, transition months, peak July).

The novel aspects of our contribution lie in the combination of classical inference with a transparent nested-model comparison and a geographic dimension. Most prior delay studies emphasize prediction accuracy on flight-level data; we instead emphasize attribution at the carrier-airport-month panel level, which is the unit at which public policy operates. The carrier-versus-weather framing of H3, in particular, is rarely tested directly in the literature even though it sits at the heart of public discourse about delays.

5.4 Limitations and Future Work

Three limitations bound our conclusions. First, the panel covers only January–August 2024; the high-volume November and December holidays are missing, so we cannot speak to year-end seasonality. Second, the DOT cause categories are reported by the carriers themselves: they can sometimes classify ambiguous delays under NAS or weather rather than under the carrier-controllable bucket, so our H3 finding is a conservative estimate of the carrier share. Third, our analytical unit is the carrier-airport-month, not the individual flight; we cannot directly model propagation between specific flights, only the aggregate footprint of late-arriving aircraft.

The most natural extensions of this work address each limitation in turn. A flight-level study using the BTS T-100 database would quantify how a delay at a hub airport cascades through the network. Joining BTS to NOAA weather observations would let us control explicitly for meteorology, providing a sharper upper bound on weather's true contribution. A mixed-effects model with random slopes by carrier would formally test the carrier-specific seasonality patterns visible in the carrier $\times$ month heatmap. Finally, extending the panel to a full calendar year would let us test whether the post-pandemic recovery patterns observed here persist into the high-volume holiday season.

5.5 Innovativeness and Contribution

Our contribution combines four classical inferential tests (ANOVA, Kruskal-Wallis, two-proportion z, Pearson/Spearman) with a nested-model comparison that quantifies driver importance, a state-level tile-grid choropleth and bubble map that surface the regional clustering of delay severity, and a Census-region ANOVA that complements the carrier and seasonal analyses with a third axis of heterogeneity. The pipeline is fully reproducible: all code, data, and figures sit in the project's GitHub repository, organized into four sequential scripts (`01_load_and_explore.py`, `02_hypothesis_tests.py`, `03_geo_and_models.py`, `04_us_maps.py`) that re-run end-to-end in well under a minute on a laptop.

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